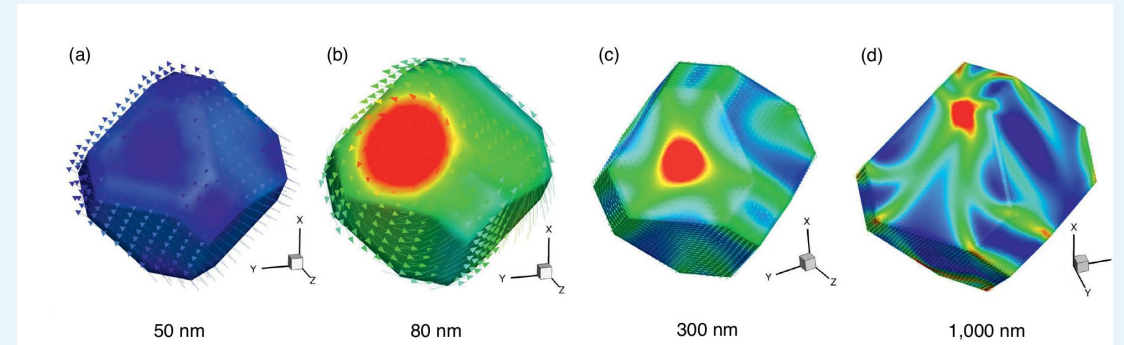


Magnetic domains in rocks

From domain walls to vortices

IRM Summer School 2026
Day 2

Tuesday July 21st, 2026, Maxwell Brown



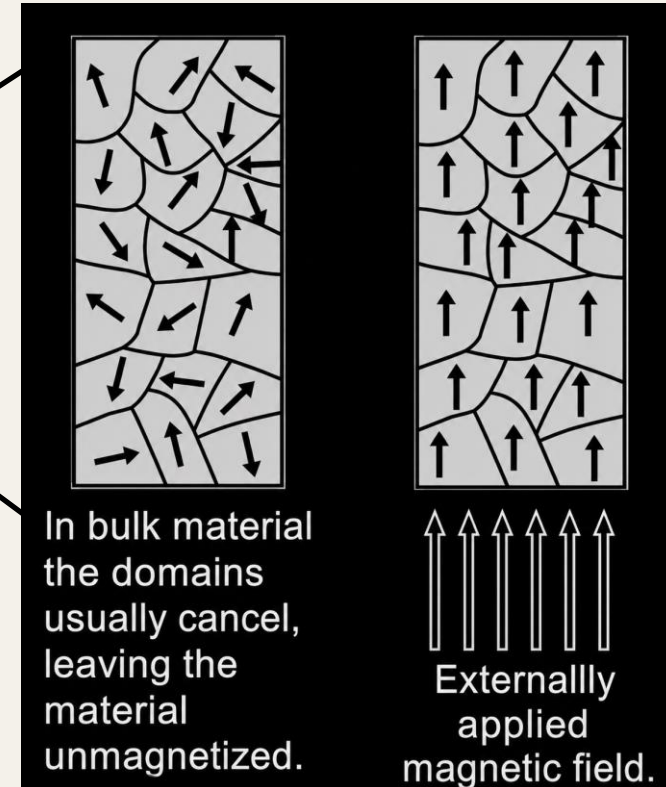
A single crystal can host a succession of non-uniform states as its size, history, and surroundings change.

Welcome to the world of domains!

Q: Why isn't the refrigerator a powerful magnet?



A: Magnetic Domains!



Spontaneous magnetization is defined as the magnetization that exists in individual regions or domains even when $B_0 = 0$. The magnetization directions of the domains may be macroscopically aligned by an external magnetic field.

Why domains?

Perhaps we can think of domains like this:

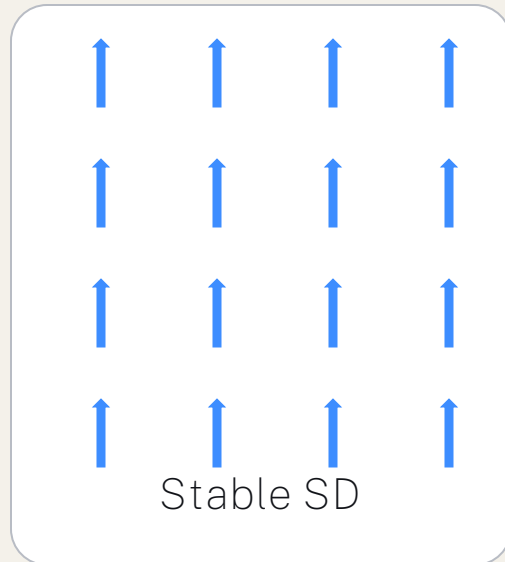
As grains grow magnetic domains allow them to reduce its total magnetic energy by dividing into regions whose magnetizations point in different directions.

A magnetic grain adopts the magnetization pattern that minimizes its total energy (not a pattern with the largest net magnetic moment).

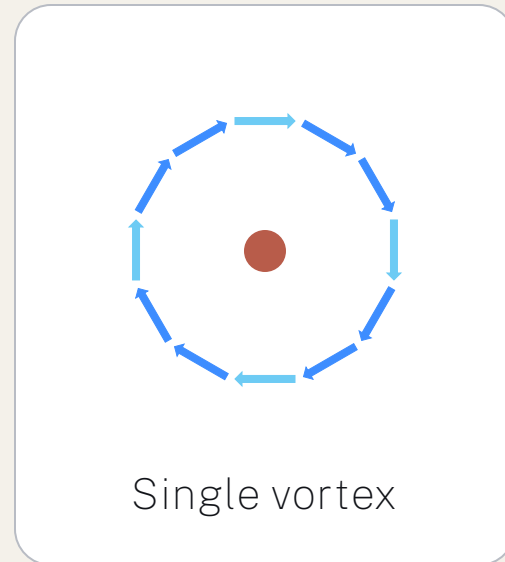
Physically small grains remain SD because making a domain wall costs more energy than it saves; larger grains can lower their total energy by forming domains, vortex states, or other nonuniform structures.

Strictly, rocks often preserve a local energy minimum — a metastable state — rather than finding the absolute lowest-energy configuration.

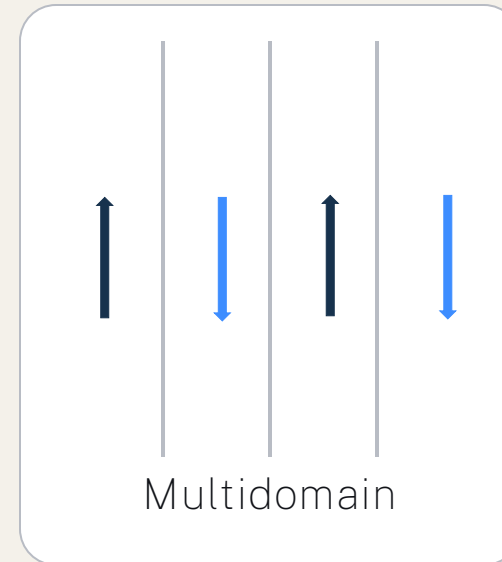
Crystals can contain several different magnetic structures



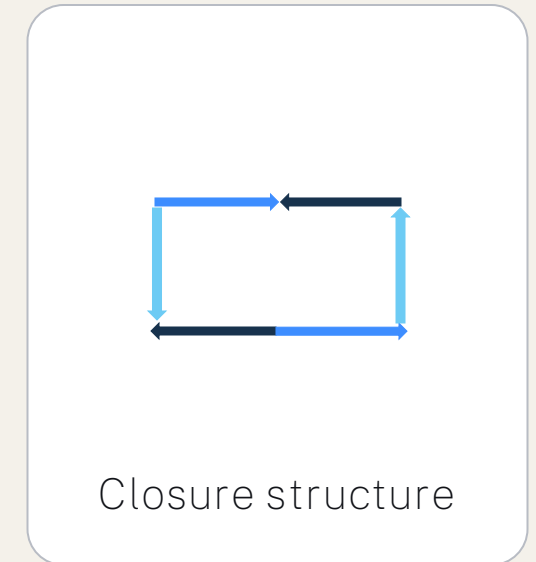
Nearly uniform
moment



Flux closes internally



Walls separate body
domains



Surface domains
reduce poles

Crystal structure does not uniquely determine micromagnetic structure.

From SD to true MD: hundreds of times wider



375 × wider

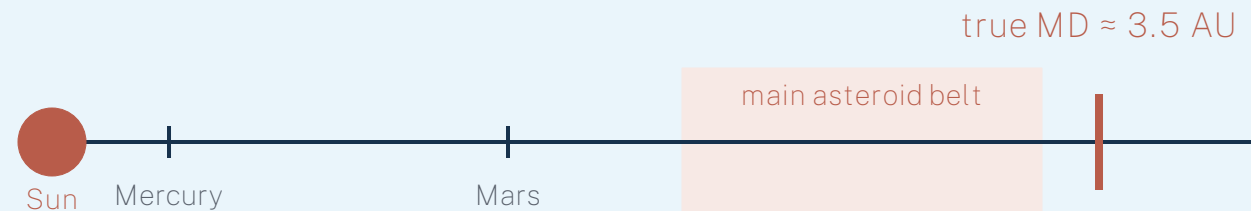
≈ 53 million × the volume
for similarly shaped grains

SOLAR-SYSTEM ANALOGY

If 80 nm were the Sun's diameter...

300 nm vortex ≈ 3.8 Suns across

30 μm true MD ≈ 3.5 AU across



Domain state controls how a rock acquires, stores, and loses remanence

More with Nick tomorrow

Acquisition

Uniform rotation, vortex nucleation, and domain-wall growth are sensitive to field and temperature.

Storage

Energy barriers, topology, and pinning determine whether remanence lasts seconds, millennia, or billions of years.

Erasure - Demagnetization

AF, thermal, low-temperature, and chemical treatments probe different reversal (demagnetization) mechanisms.

Domain interpretation is an important part of the remanence recording model and key to understanding past records of Earth's magnetic field.

Four questions!

1. What energy competition creates a domain state? (Supplementary edition)
 2. What physically occupies the historical 'PSD' interval?
 3. How do domain walls form, and structures vary?
 4. How can we observe or infer structures?
-

A domain state is the lowest accessible compromise among competing energies

$$E_{\text{total}} = E_{\text{exchange}} + E_{\text{anisotropy}} + E_{\text{demag}} + E_{\text{Zeeman}} + E_{\text{magnetoelastic}} + E_{\text{interactions}}$$

Exchange

Keep neighboring spins parallel

Anisotropy

Keep magnetization on easy axes

Demagnetizing

Reduce surface and internal poles

Zeeman

Interaction energy between moment and an externally applied magnetic field

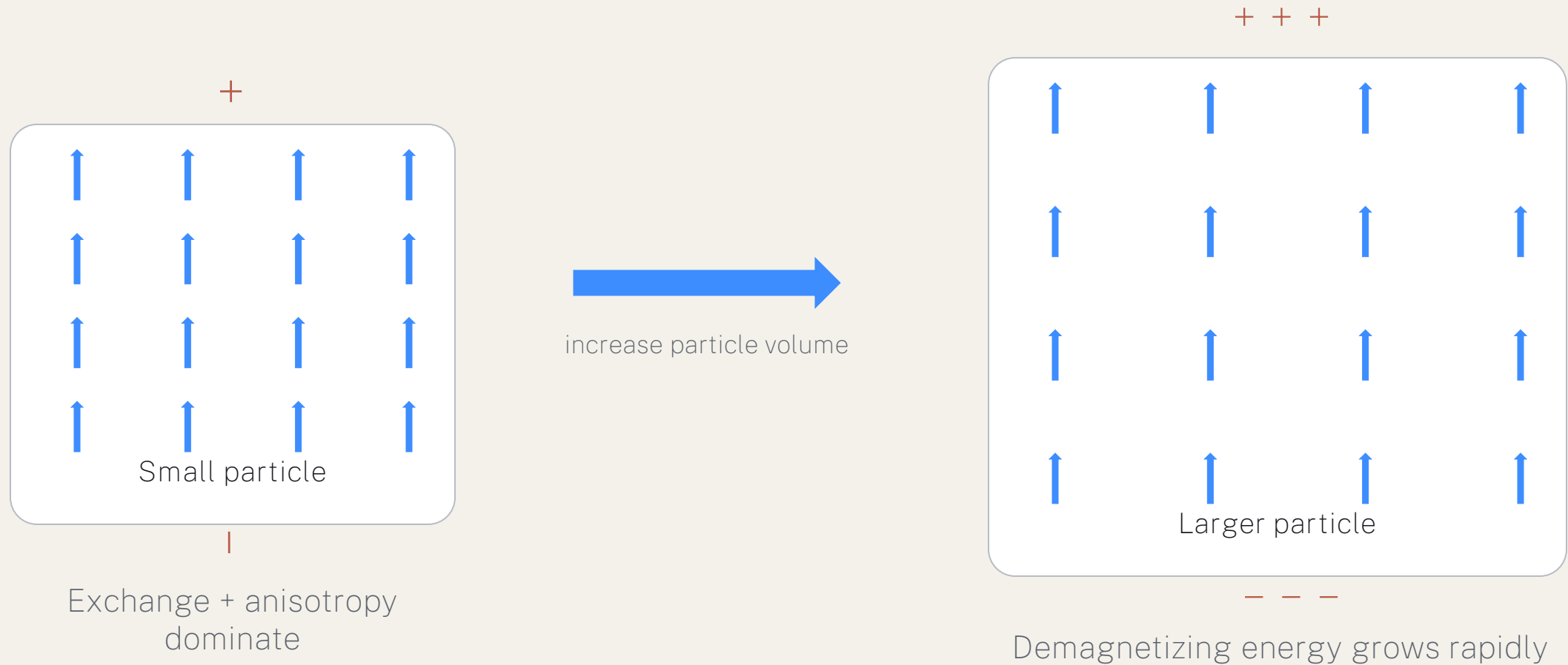
Stress + interactions

Respond to strain and neighbors

Magnetoelastic and interactions.

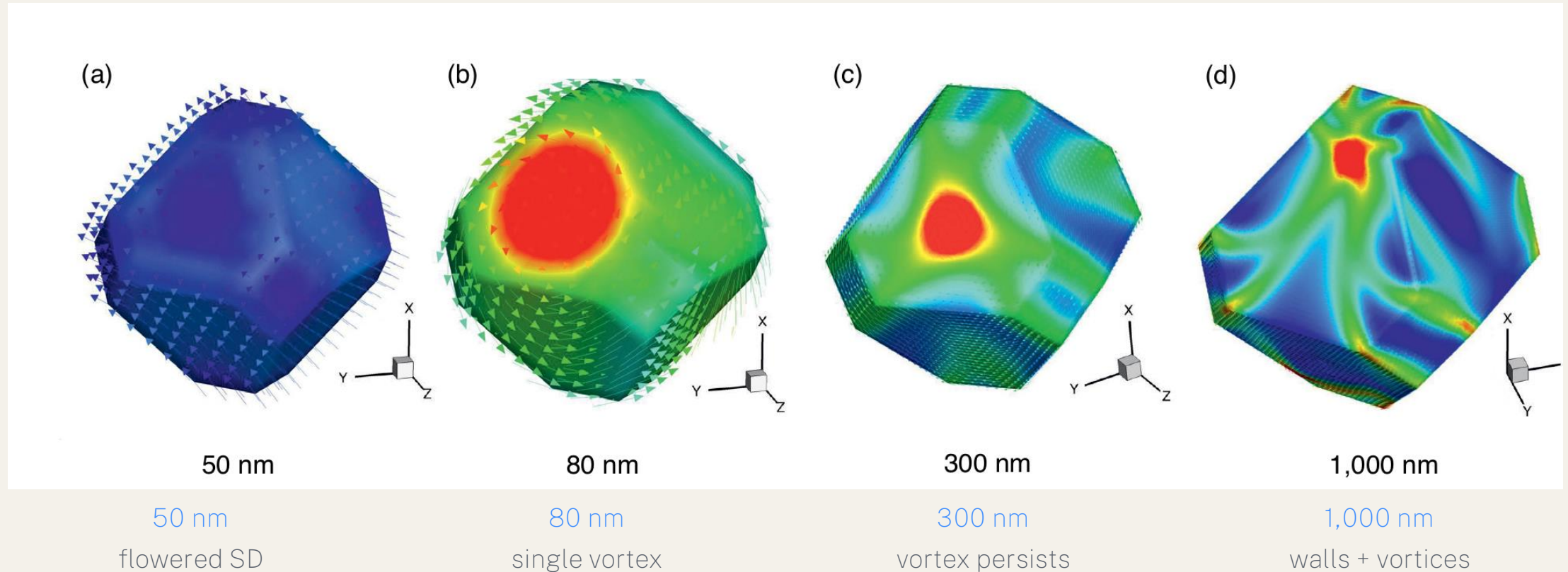
Magnetic grains do not necessarily occupy the configuration with the absolute lowest possible energy – crystals in rocks often preserve a metastable local minimum.

Surface poles make uniform magnetization progressively expensive



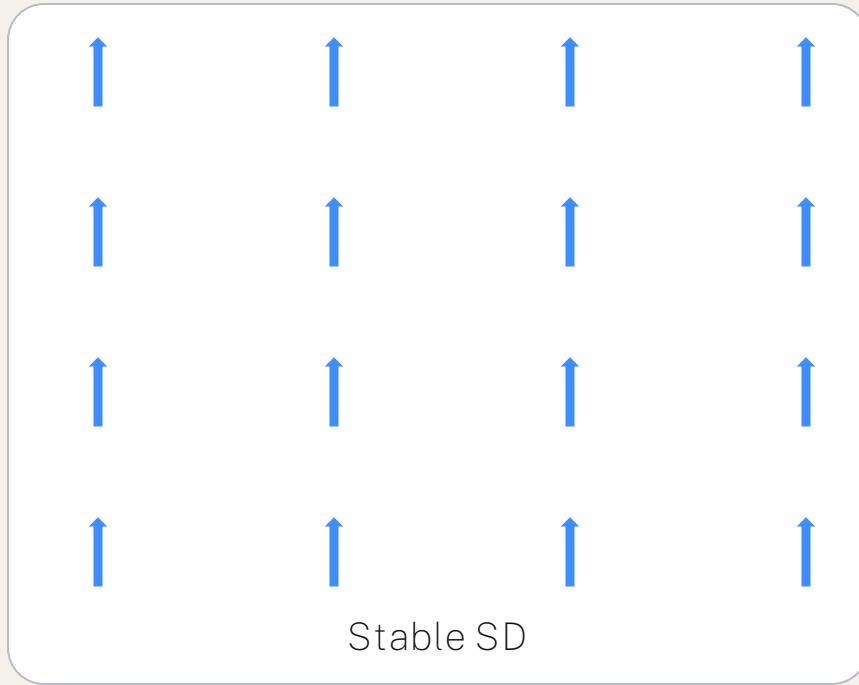
The system can lower E_{demag} by flowering, curling into a vortex, or dividing into domains.

Increasing size produces a sequence — not a single SD-to-MD jump



Non-uniformity begins before walls; vortices can survive after domain walls appear.

Stable single-domain behavior is a good starting point



one dominant magnetic moment

No internal wall

Moment reversal requires rotation, buckling, fanning, or curling rather than translating an existing wall.

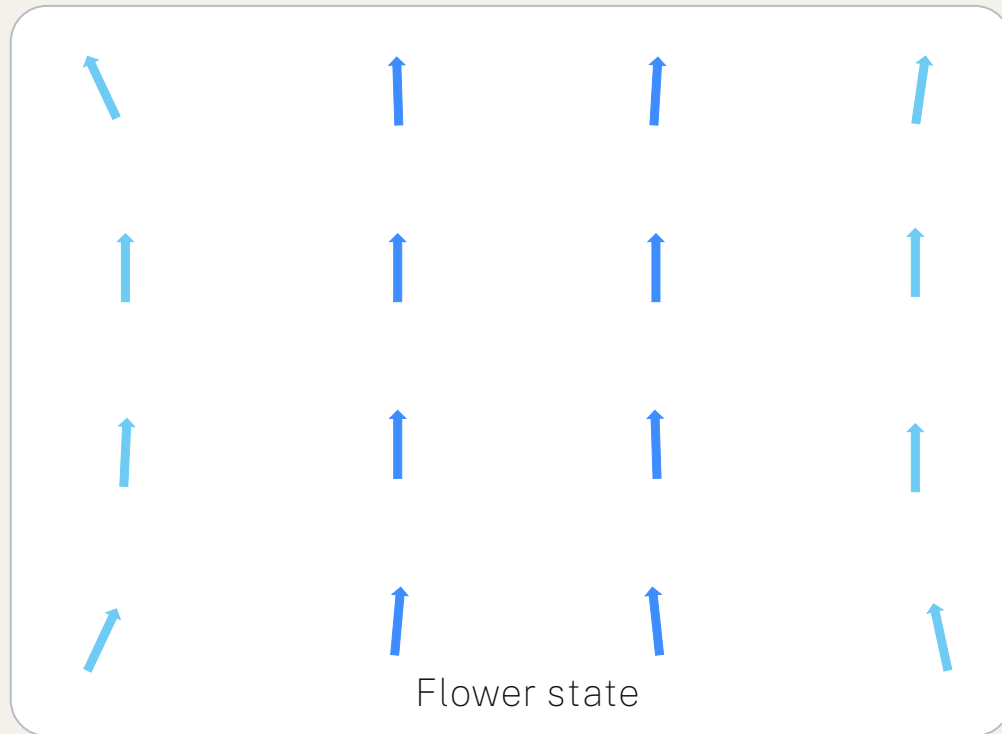
Large remanence

Random ideal uniaxial SD particles can retain high saturation remanence relative to M_s .

Finite relaxation time

Stability is an energy barrier observed over a timescale — not an absolute property of the particle.

Large SD particles already 'flower' at their margins



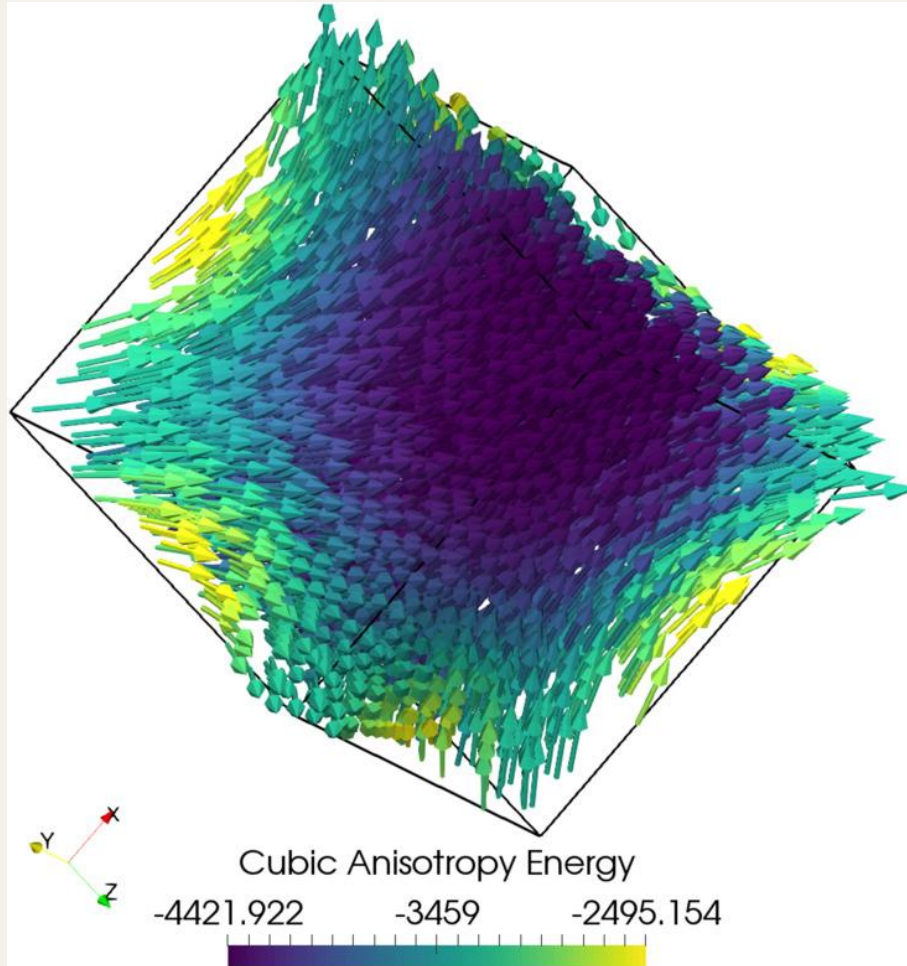
edge moments deflect toward the surface

$\approx 92\%$

of saturation remanence in a modeled
large flower-state magnetite particle

- Still one connected magnetic state
- No classical domain wall
- Edge deflection reduces surface poles
- The precursor to vortex nucleation

Large SD particles already 'flower' at their margins

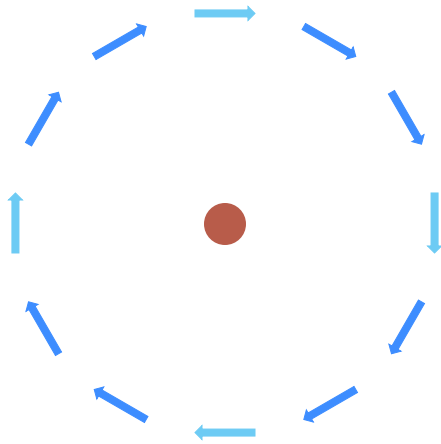


$\approx 92\%$

of saturation remanence in a modeled large flower-state magnetite particle

- Still one connected magnetic state
- No classical domain wall
- Edge deflection reduces surface poles
- The precursor to vortex nucleation

A vortex closes flux by curling around a narrow core



Single vortex

circulation minimizes external stray field
most moments curl around the core

Chirality

Clockwise or anticlockwise circulation around the core.

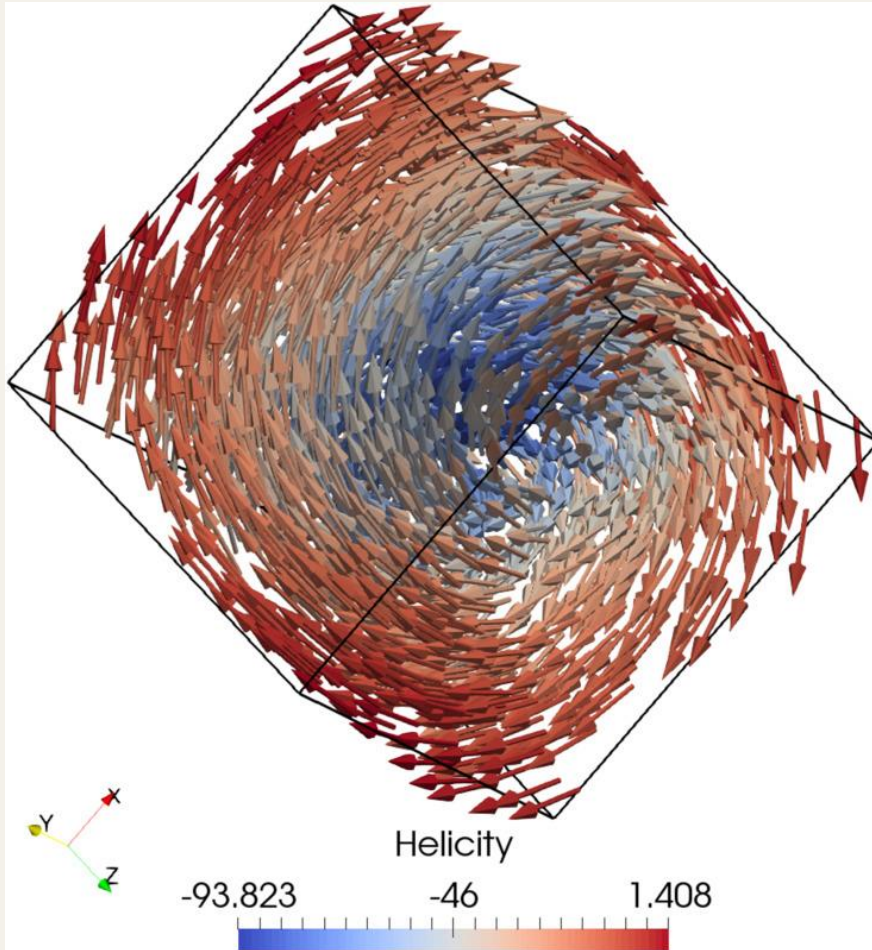
Core polarity

The core magnetization points out of or into the particle plane.

Net remanence

Not necessarily zero: shape, core position, defects, and history break perfect cancellation.

A vortex closes flux by curling around a narrow core



Chirality

Clockwise or anticlockwise circulation around the core.

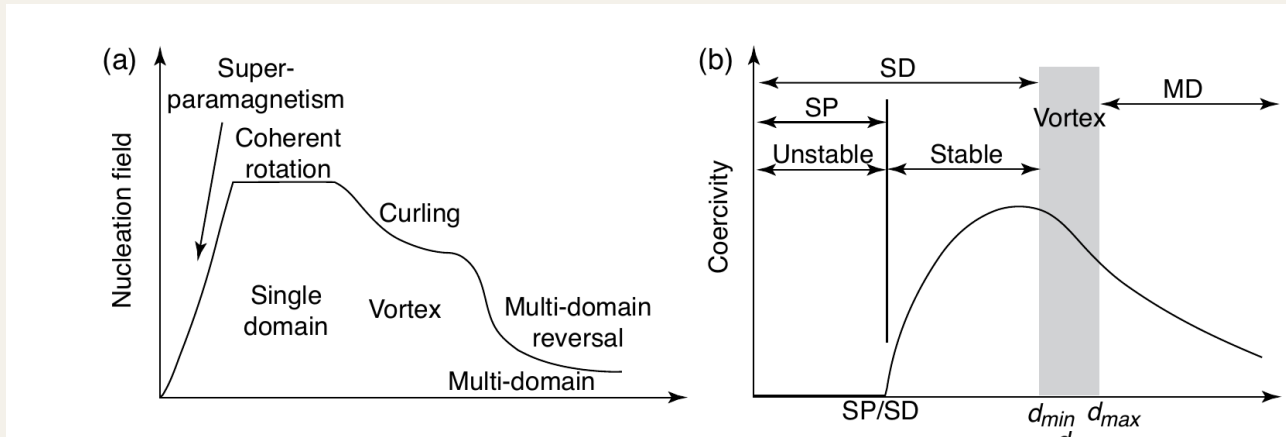
Core polarity

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Not necessarily zero: shape, core position, defects, and history break perfect cancellation.

Vortices reverse through nucleation, motion, and annihilation



Distinct nucleation and annihilation fields can imprint shoulders or jumps on a hysteresis loop.

1 Saturated state
moments largely follow H

2 Vortex nucleates
curling reduces stray field

3 Core moves
defects and shape guide the path

4 Vortex annihilates
opposite near-saturation state

'PSD' was coined to name the broad SD-like-to-MD-like transition

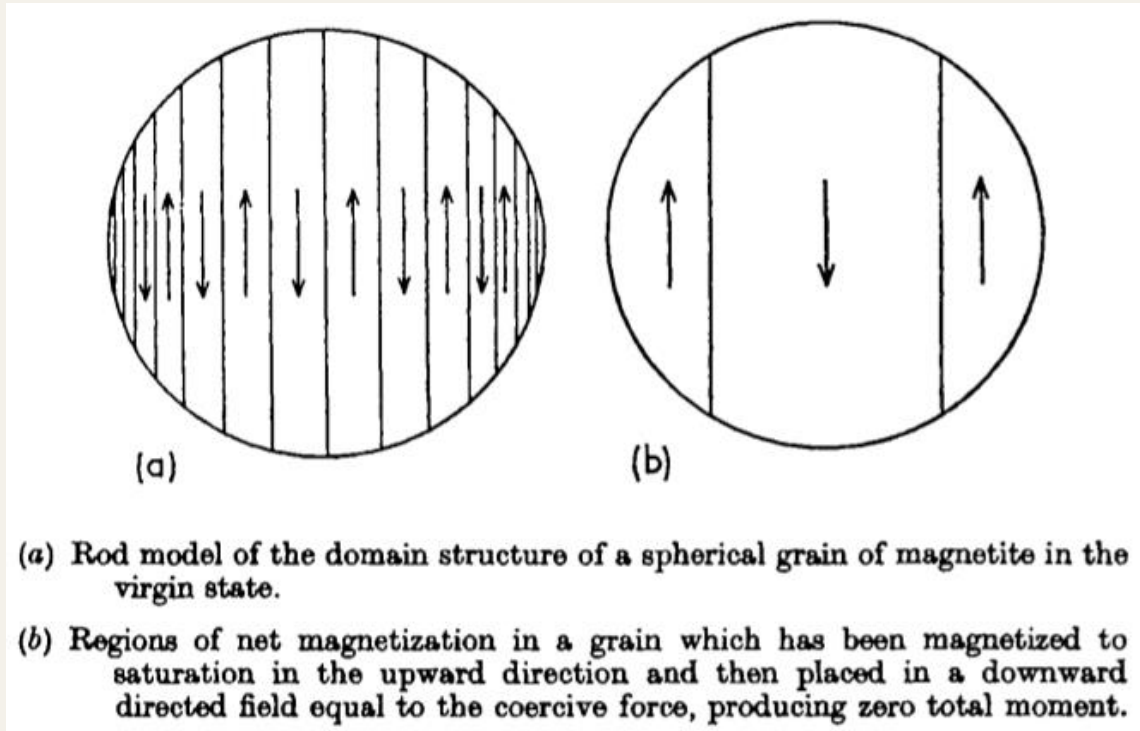
Stacey (1961) phrased his PSD observation as:

“[PSD] magnetic properties may be substantially different from true multidomain behaviour and more like a superposition of single-domain properties upon the multidomain character. The essential feature is the appearance of permanent dipole moments which can be reversed [by AF] but not destroyed by demagnetization, and the effect is therefore termed pseudo-single-domain behaviour, which can occur in grains which are much too large to be true single domains”

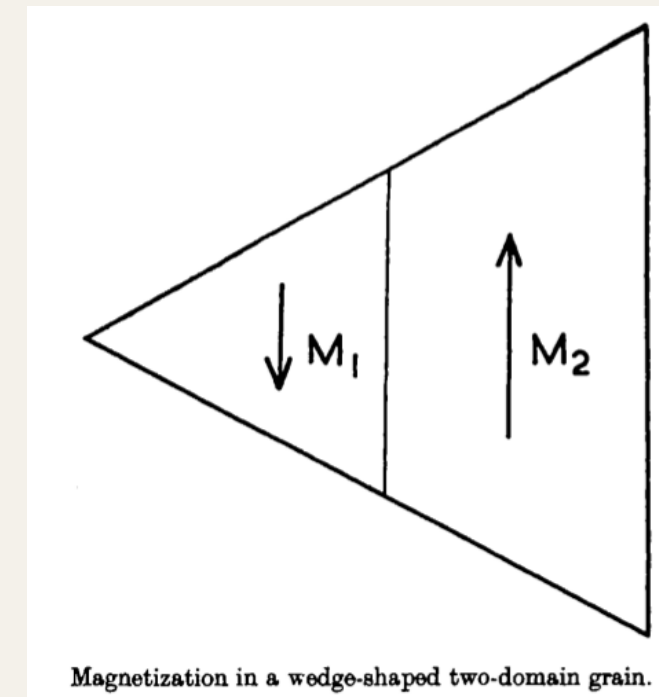


The observation was correct: properties change gradually.

Stacey's initial model



Perfect MD grains will have no net magnetization, as the moments in adjoining domains cancel (like for our fridge).



Wedge

- Balance of wall energy and magnetostatic energy due to polarity of surface at the end of domains.
- Imbalance of moments (net moment up).
- A stray dipole field is produced.

Explanation (now superseded by Vortex state)

- Natural MD grains are usually irregular in shape.
- Small irregularities in larger grains cause a departure from perfect domain configurations.

Why the fuss?

Day et al. (1977) Diagram

- Hysteresis parameters used in combination to understand domain state.
- Remanence/Infield for magnetization and coercivity.
- Defined boundaries for domains – PSD is one.
- Found in over 2500 studies.
- Over time we have realized that the majority of magnetite or titanomagnetite bearing samples fall in the PSD box.

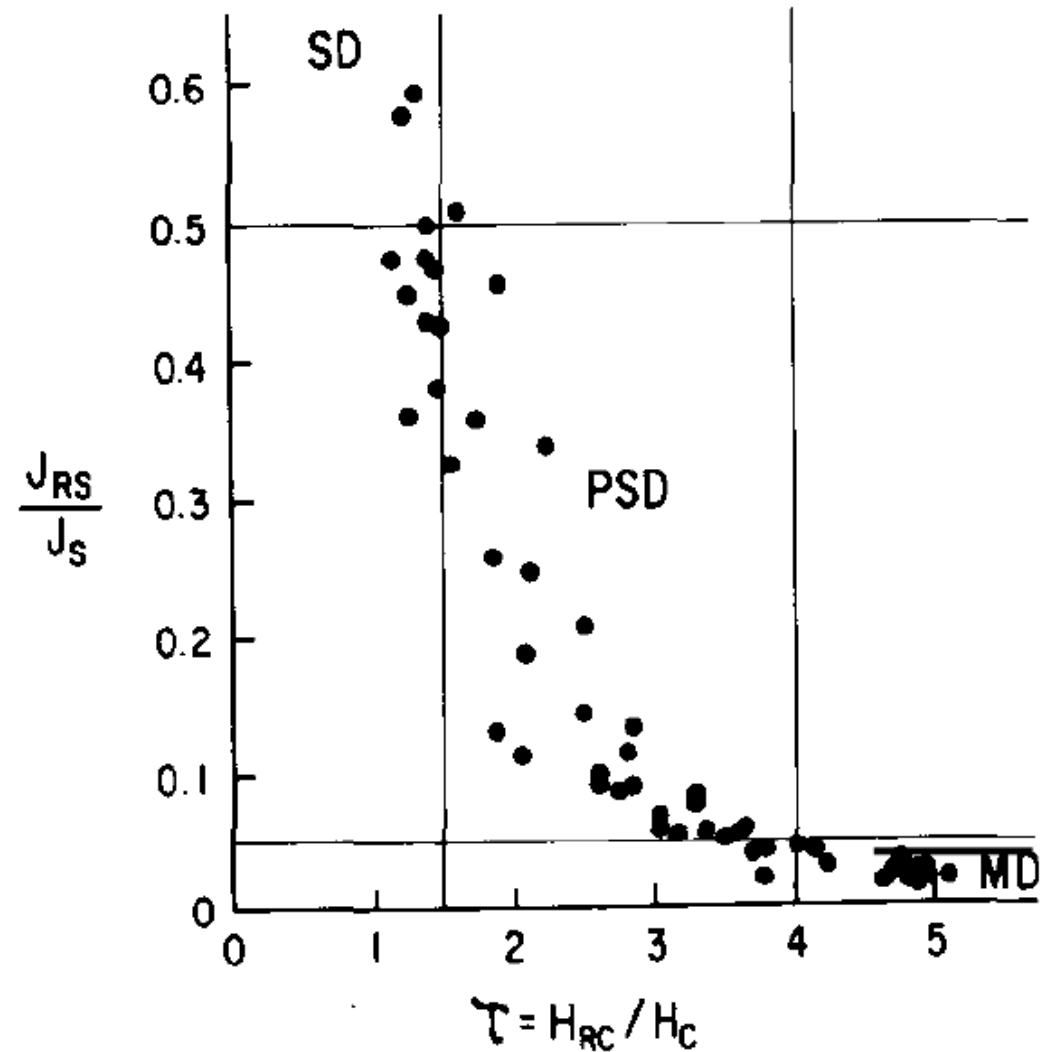
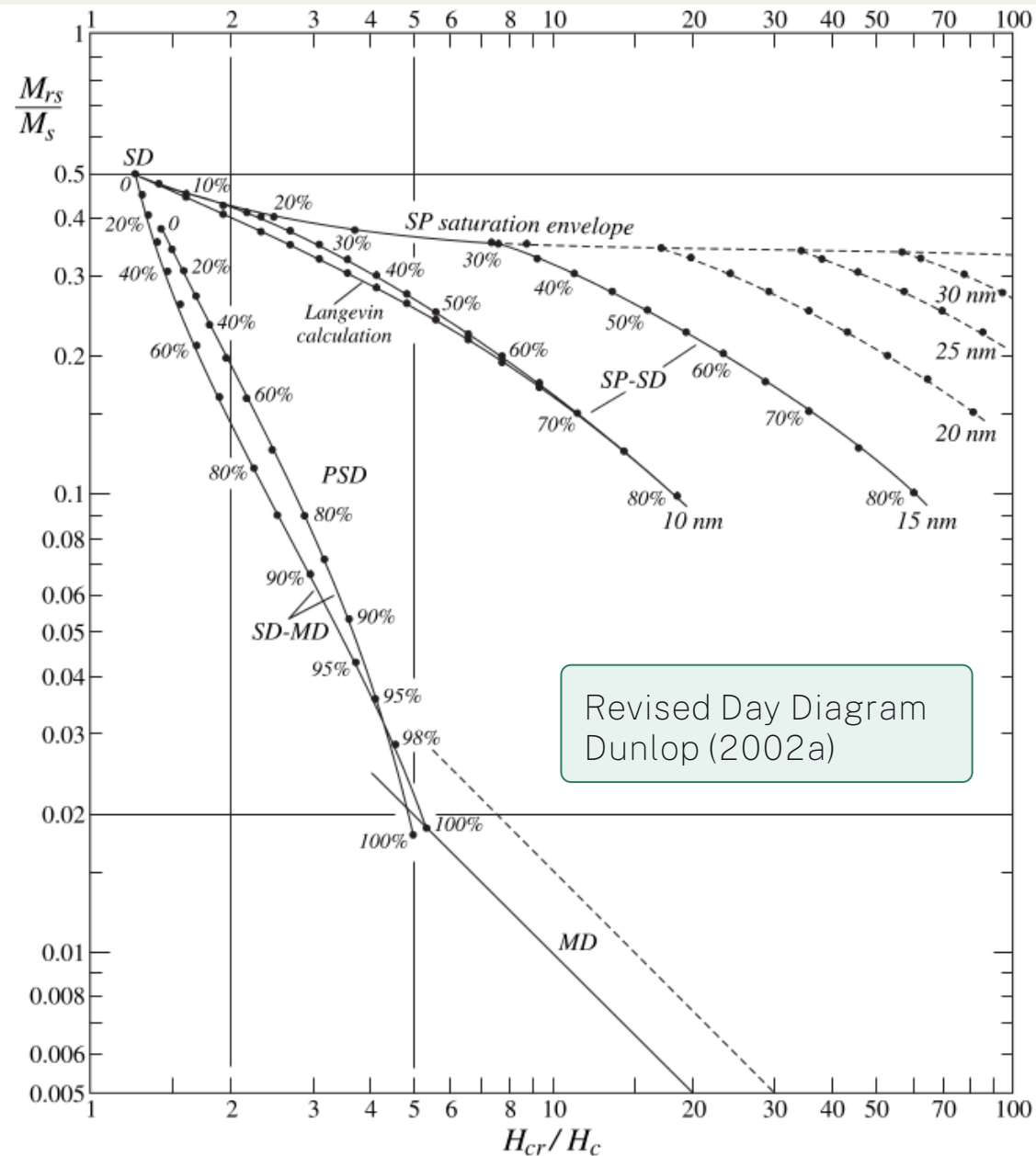


Fig. 4. J_R/J_S versus H_{RC}/H_C .

Why the fuss?

Day et al. (1977) Diagram

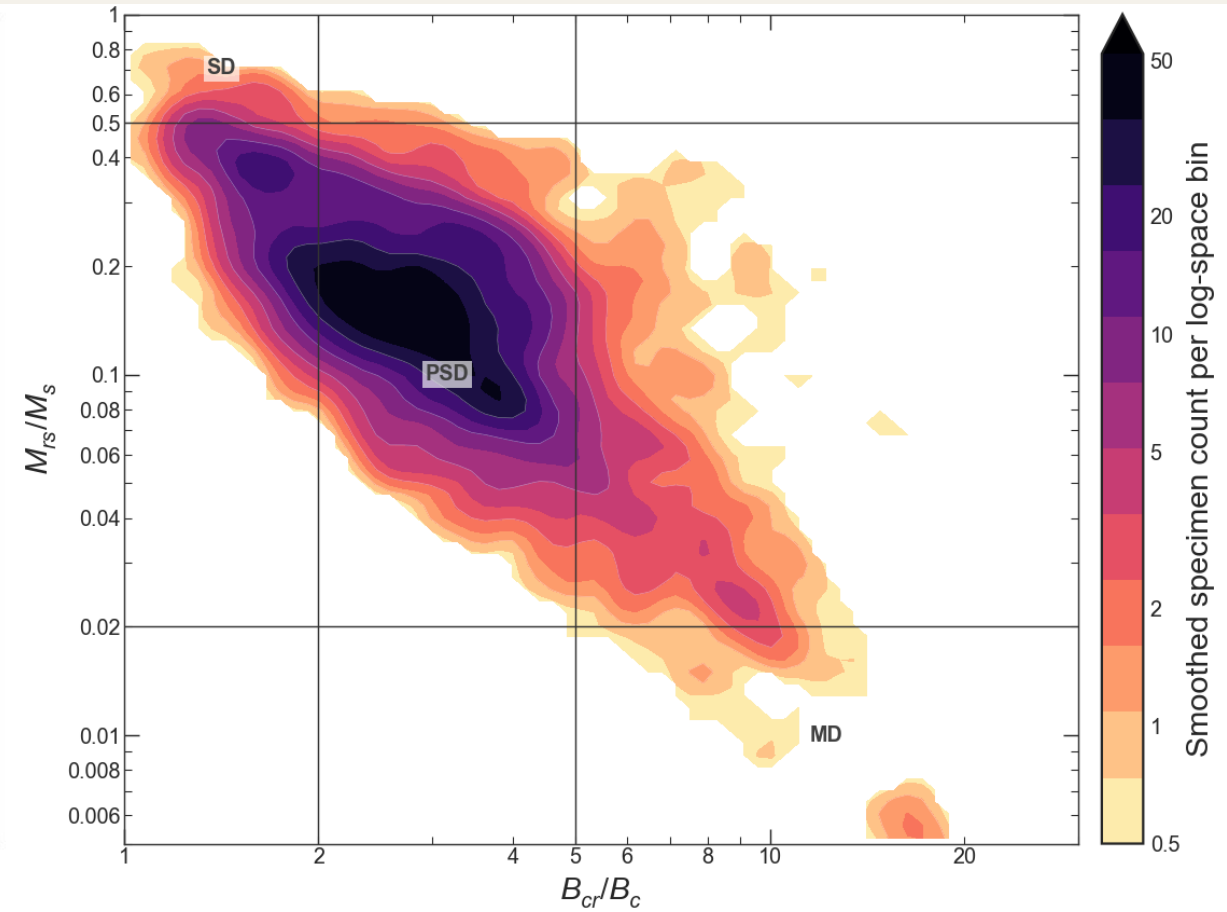
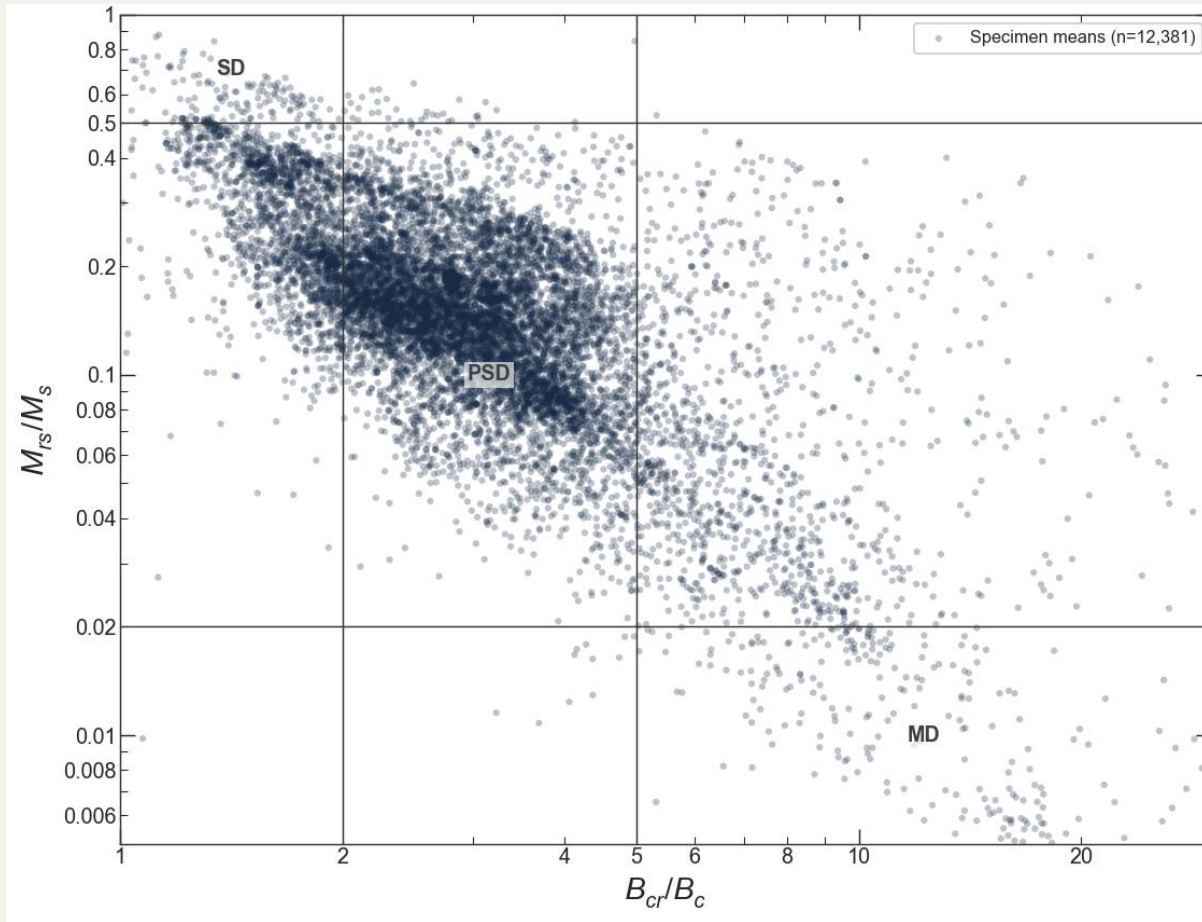
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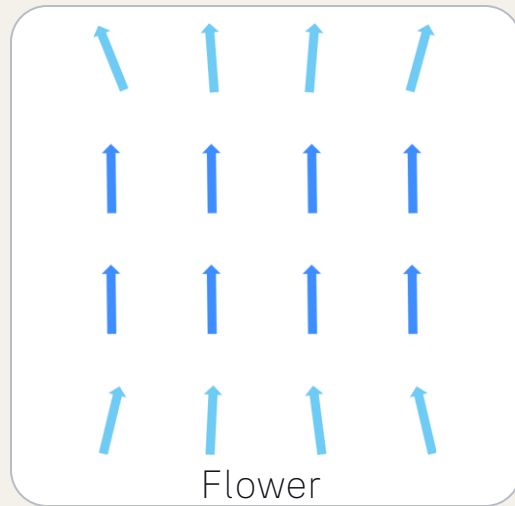
Why the fuss?

IRM database ratios

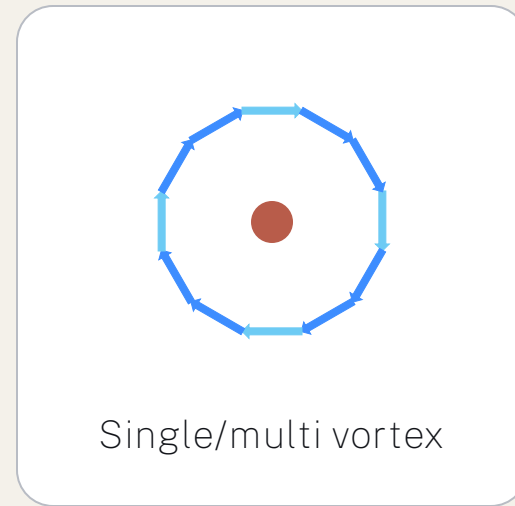
Room temperature measurements with $Q_f > 2$
12,381 specimens



A bulk 'PSD' signature can arise from several different structures

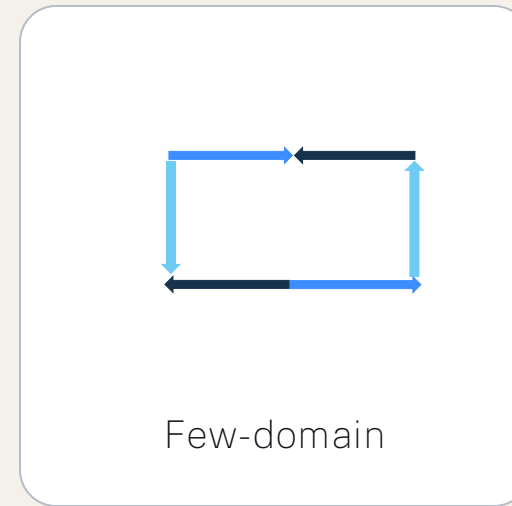


Flower
large SD



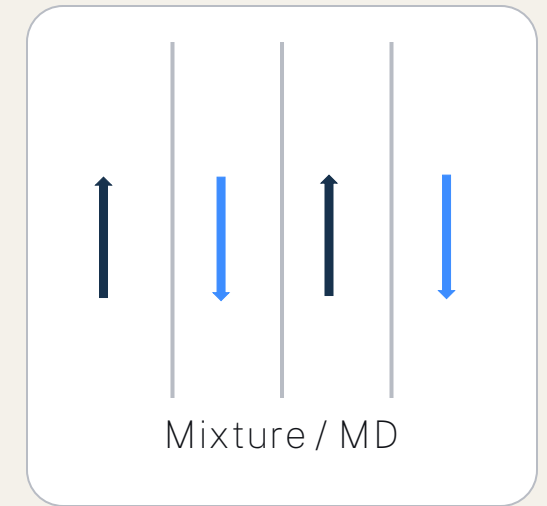
Single/multi vortex

curling state(s)



Few-domain

closure + wall



Mixture / MD

size + interactions

Day plot: similar M_{rs}/M_s and B_{cr}/B_c ratios do not prove identical internal magnetization.

Roberts' update: PSD is not a distinct physical domain state

~~PSD~~

Not a unique micromagnetic
configuration

PSD region defined by complexity

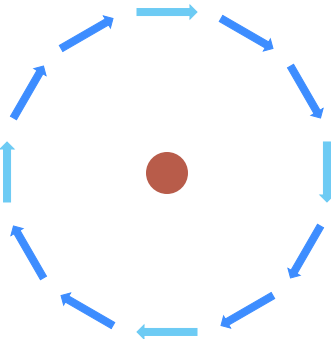
Flower, single vortex, multi-vortex, describe actual magnetization structures.

Keep 'PSD-like' as a cautious phenotype

Bulk ratios may occupy the historical PSD field without identifying which structure or mixture produced them.

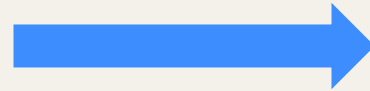
A starting point for further rock magnetic exploration.

True multidomain behavior begins when walls become energetically cheaper

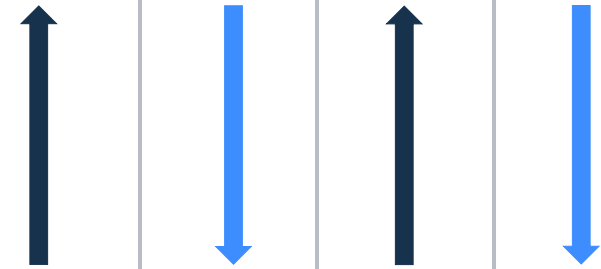


Flux closure by curling

exchange cost is distributed continuously



increasing size

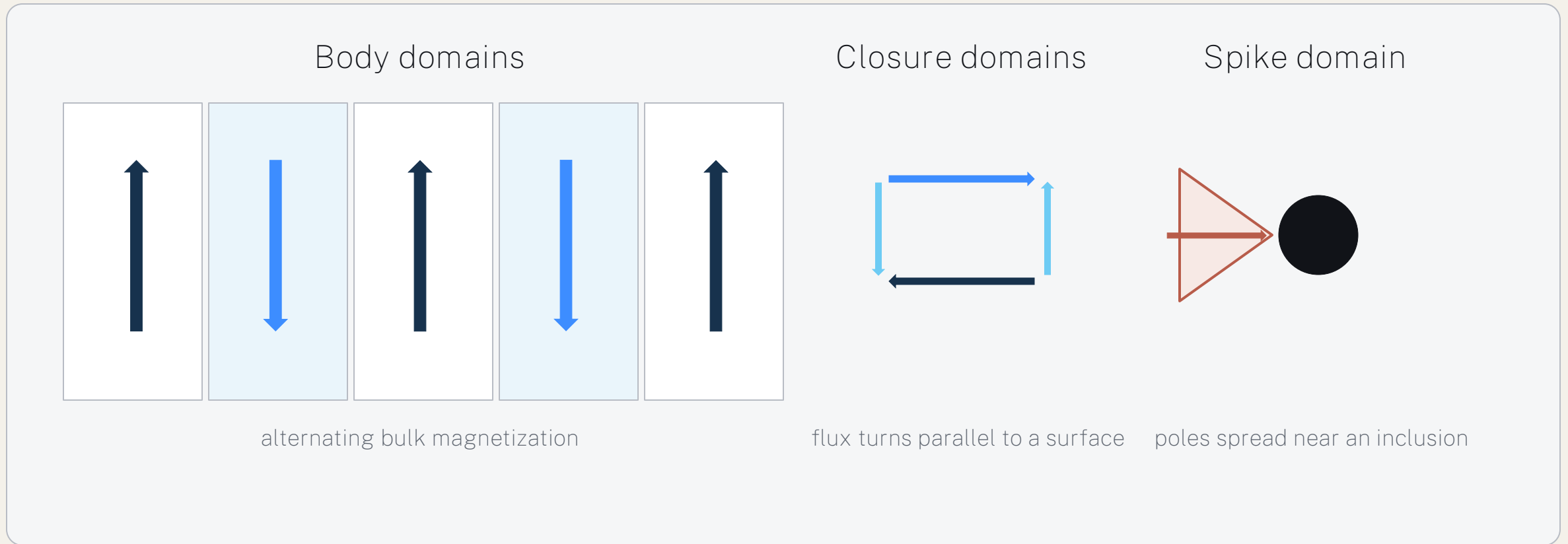


Flux closure by subdivision

wall energy is paid at discrete boundaries

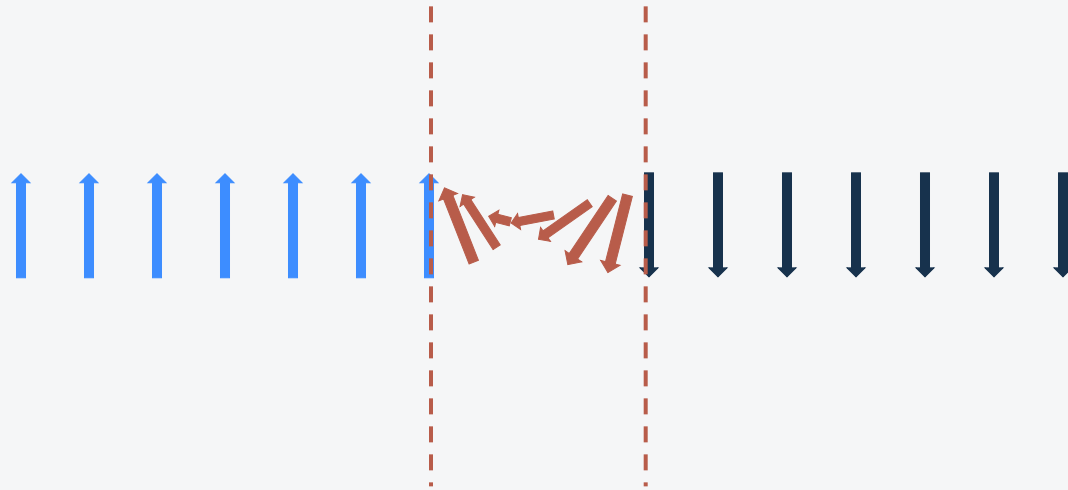
At sufficiently large volume, adding walls reduces demagnetizing energy more than it costs in wall energy.

Body, closure, and spike domains reduce different sources of stray field



Real grains combine these structures in 3D

A domain wall: exchange-anisotropy compromise - not a sharp boundary



Bloch Wall: domain wall width δw

Exchange favors width

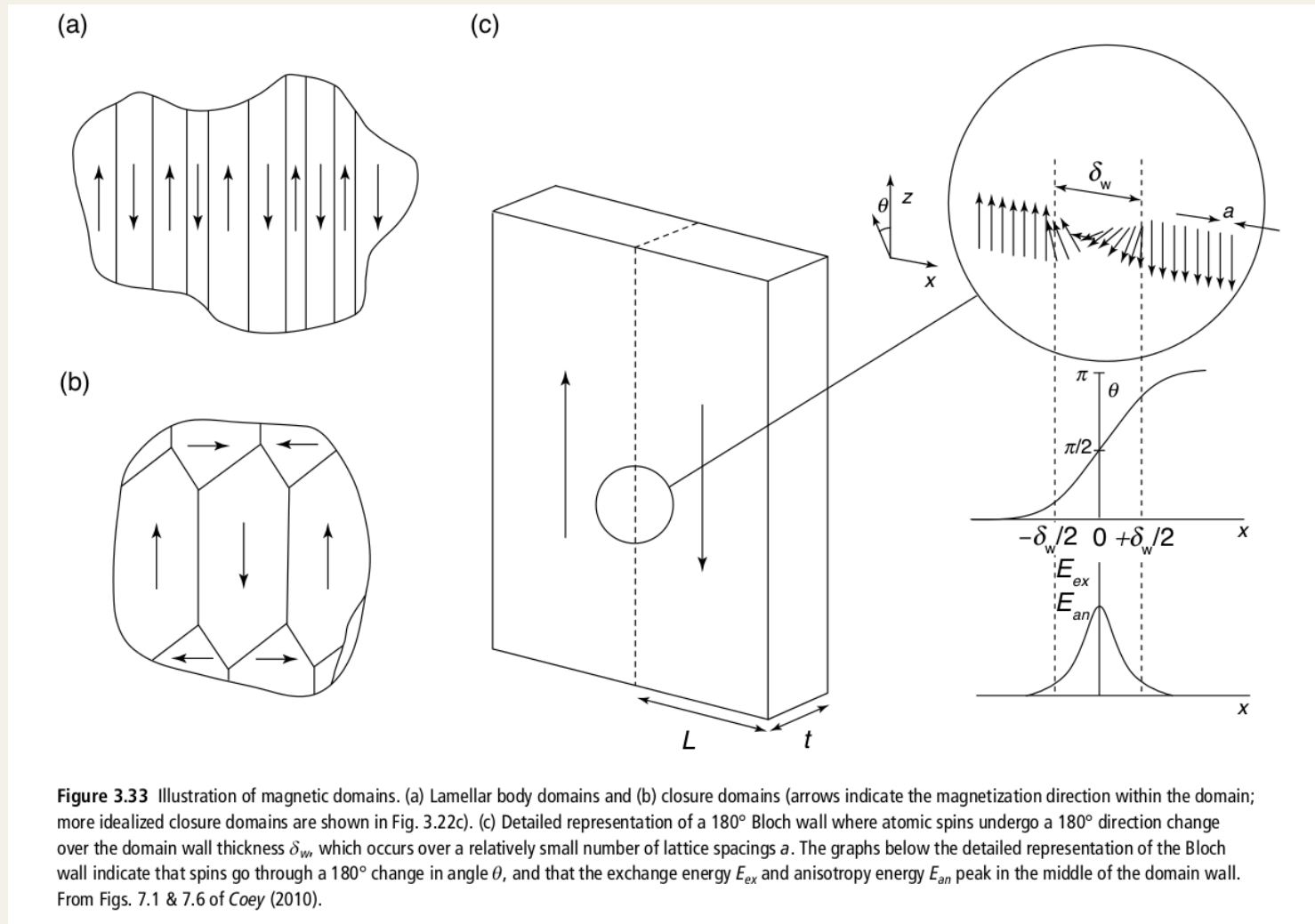
Slow rotation keeps neighboring spins nearly parallel.

Anisotropy favors thinness

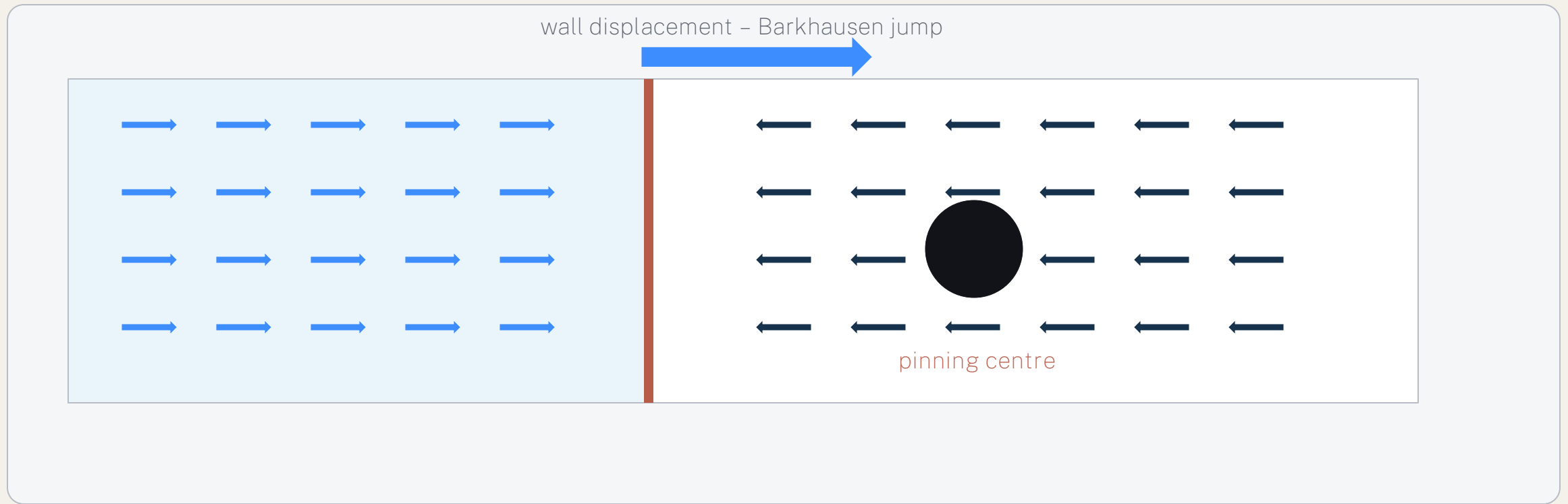
A narrow wall limits the volume away from easy directions.

Bloch Wall: boundary between magnetic domains, across which the magnetization rotates smoothly rather than reversing abruptly

Domain walls



Wall motion is easy until defects, stress, or inclusions pin it



reversible bulging

irreversible jump

new wall nucleation

annihilation

What does this look like through observations?

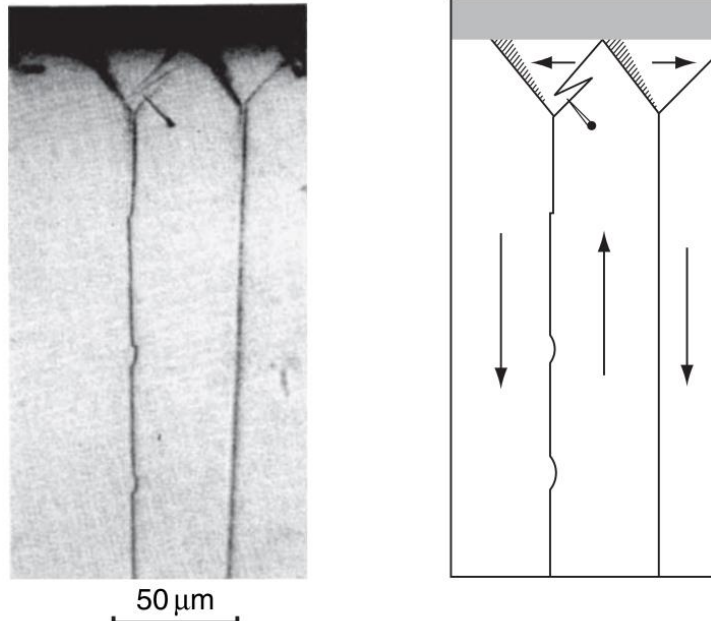


Figure 8 Lamellar body domains and wedge-shaped surface closure domains imaged by the Bitter colloid method on a $\{110\}$ surface of a single crystal of magnetite. Modified from Özdemir Ö, Xu S, and Dunlop DJ (1995) Closure domains in magnetite. *Journal of Geophysical Research* 100: 2193–2209.

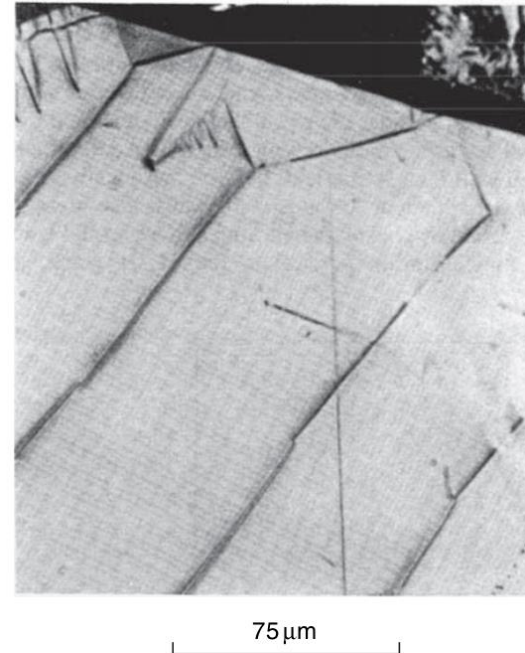


Figure 9 Closure domains observed on a $\{110\}$ surface in a magnetite single crystal. The boundary (top) is a deep crack. The body and closure domains are magnetized along $[\bar{1}11]$ and $[1\bar{1}1]$ easy directions, respectively. The 70.5° and 109.5° walls bisect the angles between easy axes, as predicted. Modified from Özdemir Ö, Xu S, and Dunlop DJ (1995) Closure domains in magnetite. *Journal of Geophysical Research* 100: 2193–2209.

MD remanence is real — but more history-dependent than ideal SD remanence

Low-field susceptibility

Often high because walls can move or bulge with little rotation of the full magnetization.

Remanence

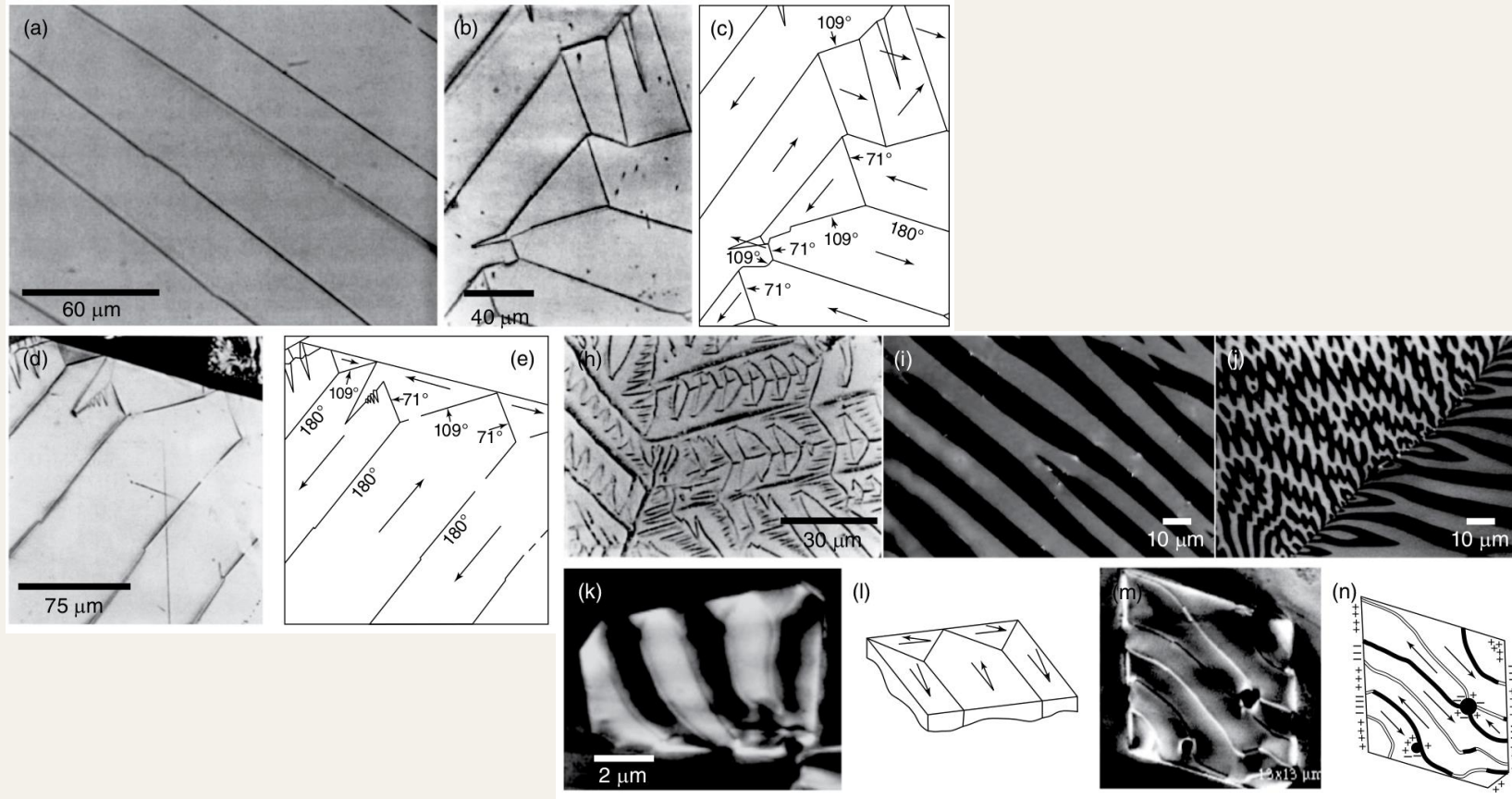
Often lower than ideal SD because neighboring domains cancel, but pinned walls and metastable states preserve a small net moment.

Stability

Controlled by wall nucleation and pinning barriers; grain history can matter as much as equilibrium structure.

'MD' describes a reversal mechanism family — not necessarily incapable of carrying remanence.

Real magnetite domains are a micromagnetic jungle



Surface cut

A polished plane intersects a three-dimensional structure.

Technique contrast

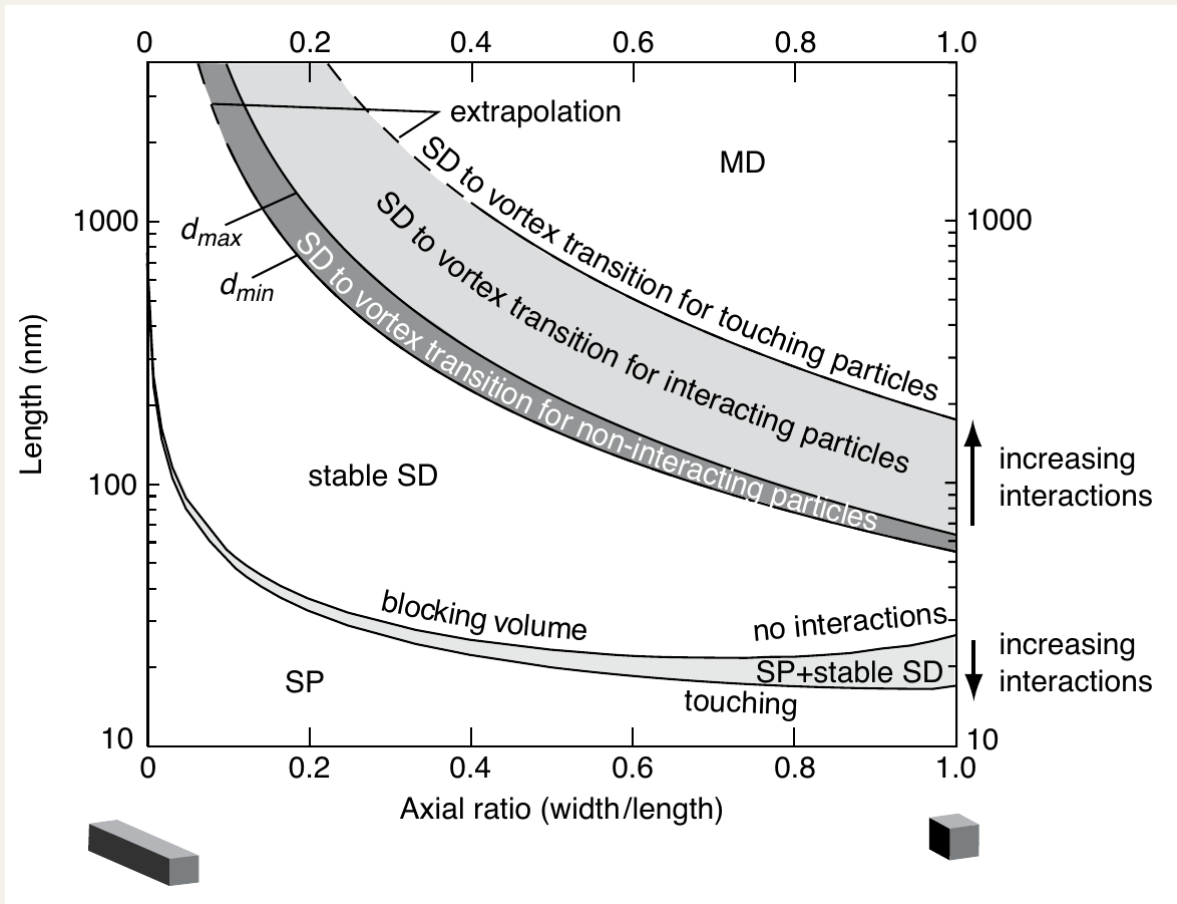
Bitter, Kerr, and MFM respond to different aspects of stray field or magnetization.

Microstructure

Cracks, inclusions, stress, and crystallography reshape walls and closure patterns.

Yiming will revisit this structure on Monday with micromagnetic modeling

Magnetite's narrow SD range opens into a broad vortex transition



Shape

Elongation strengthens shape anisotropy and expands stable SD behavior.

Interactions

Neighboring particles shift blocking and vortex-transition boundaries.

History

Different d_{min} and d_{max} values reveal hysteresis between vortex nucleation and persistence.

Shape and interactions move domain thresholds

Elongation

increases shape
anisotropy

stable SD extends larger

Equant shape

weakens shape
anisotropy

vortex nucleates earlier

Dense chains

add collective anisotropy

particles can reverse
together

Random clusters

create local interaction
fields

thresholds broaden and
split

No mineral-only critical sizes, more that boundaries belongs to a particle system.

Vortex magnetite can be a robust ancient recorder

10^9 years

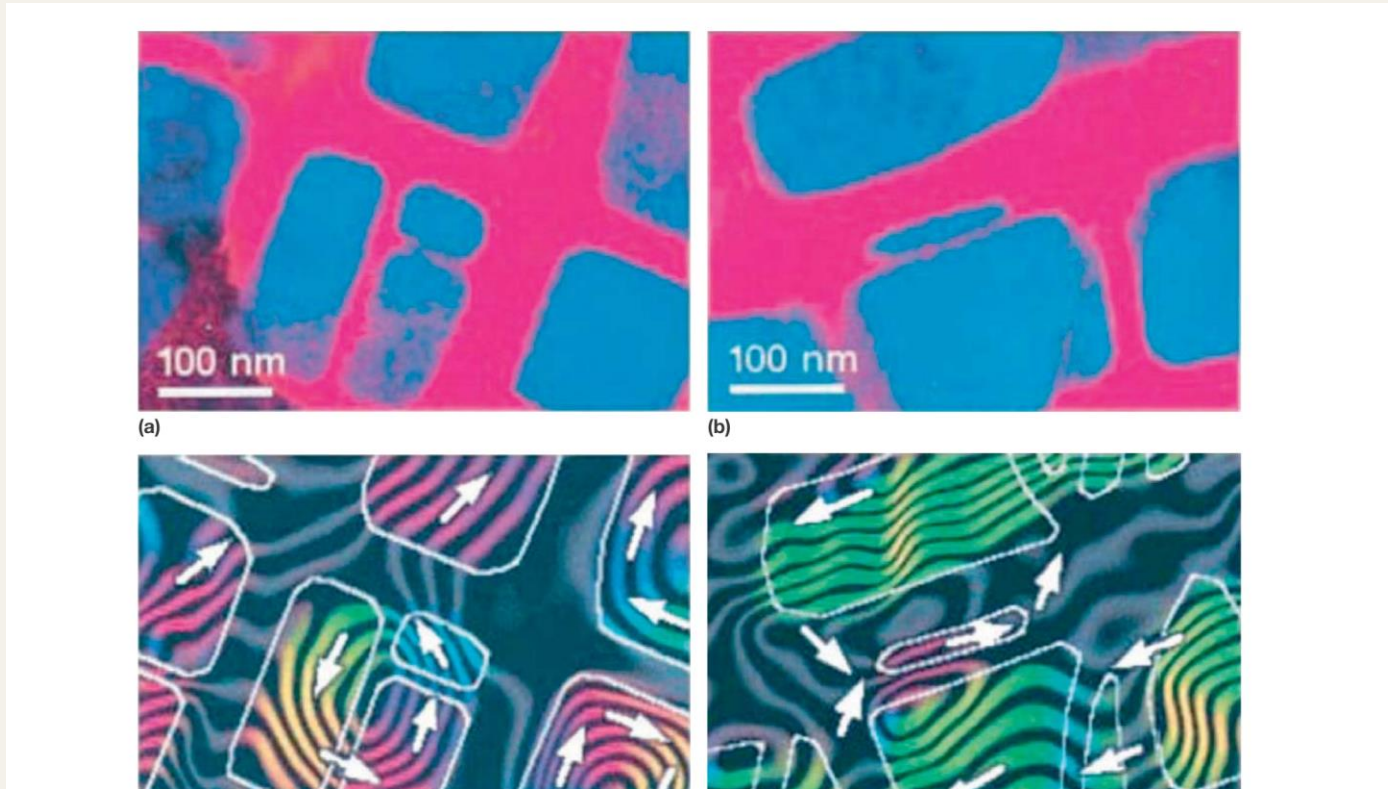
stability calculated to be possible for suitable vortex-state magnetite.

More with Nick tomorrow.

- Peak thermal stability occurs near the upper flower-state size in some models
- Vortex topology provides a substantial reversal barrier
- Low external stray field does not mean zero or unstable remanence
- Particle shape, defects, and interactions still control the actual lifetime

Paleomagnetic quality should be evaluated by stability and recording fidelity — not by whether a point falls in the old PSD field.

Exsolution can couple many particles into a magnetic superdomain



Individual blocks

Some magnetite regions are single-domain; others contain vortices.

Flux crosses the gap

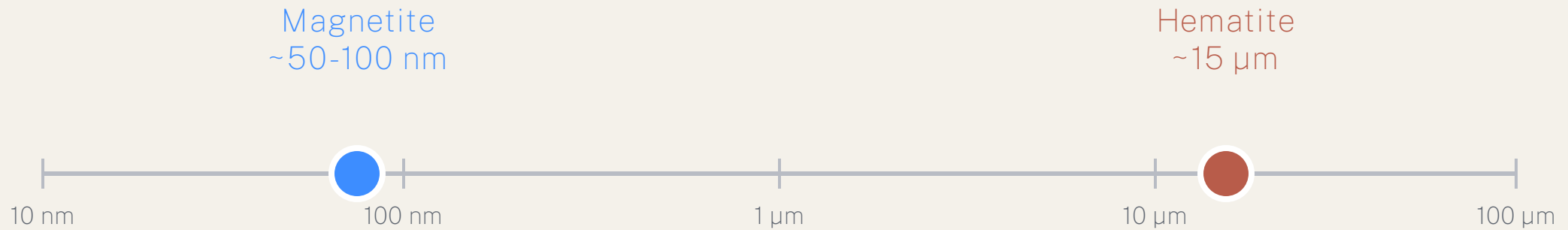
Magnetostatic fields link neighboring blocks across paramagnetic ulvöspinel.

Collective unit

The relevant recorder can be a supervortex or superdomain spanning several crystals.

Hematite's weak magnetization postpones subdivision to much larger sizes

Approximate equidimensional SD upper guide



Low M_s

Much weaker demagnetizing energy than magnetite at the same volume.

Strong anisotropy

High coercivity and large barriers can stabilize remanence.

MD can still record

Large hematite domains may preserve long-term NRM despite walls.

No imaging method sees the whole three-dimensional magnetization

Method	Best at seeing	Main blind spot
Bitter patterns	surface poles and walls	coarse grains; surface preparation
Kerr microscopy	surface magnetization changes in field	optical resolution; orientation sensitivity
MFM	stray-field gradient above a surface	tip convolution; weak-flux vortices can hide
Electron holography	projected in-plane magnetic induction	thin sample; 2-D projection ambiguity
3-D X-ray methods	vortices, walls, and topology in volume	limited access, resolution, and throughput

Combine image contrast with crystallography, topography, and a forward model of what the instrument measures.

“Bulk” measurements constrain domain behavior, but rarely identify one structure

Hysteresis + Day plot

Useful for remanence and coercivity ratios, trends, and mixtures. Intermediate ratios are non-unique and should not be translated directly into a micromagnetic state.

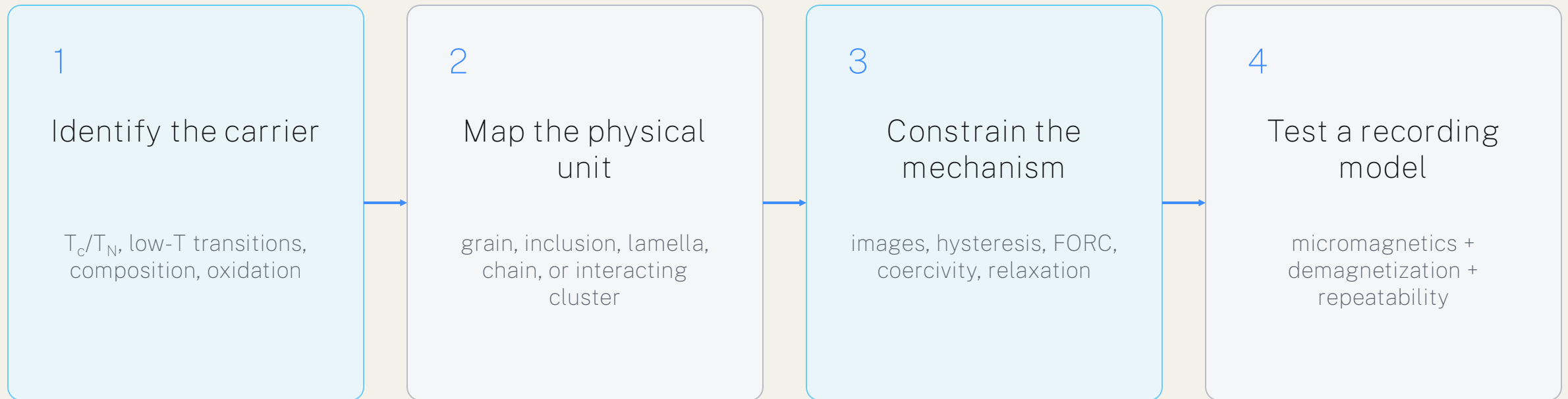
FORC diagrams

Resolve distributions of irreversible switching and interaction fields. Ridges, lobes, and vertical spreading constrain processes but still require mineralogical and micromagnetic context.

Temperature + demagnetization

Low-temperature, AF, and thermal responses reveal unpinning, relaxation, transformations, and stability. Chemical alteration can mimic or erase domain effects.

Interpretations combine mineralogy, texture, and magnetic behavior



Suggestion: report observations — not just 'PSD' or 'MD'.

Key ideas to remember during interpretation

1. Domain states are energy minima (not grain-size labels).
 2. Flower and vortex states physically occupy much of the old PSD interval.
 3. Vortex magnetite can be an exceptionally stable recorder.
 4. True MD behavior is governed by wall nucleation, motion, and pinning.
 5. Mineralogy, texture, interactions, and history influence domains.
-

References

Roberts, A. P. (2025; online 2026). Mineral Magnetism. Cambridge University Press. Chapter 3, pp. 31-71. doi:10.1017/9781108935630.005.

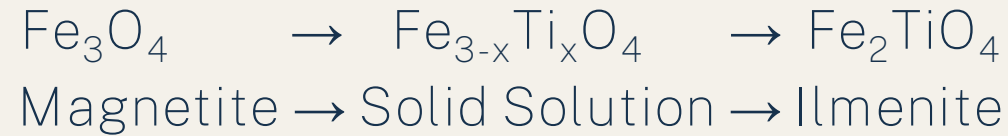
Harrison, R. J., Dunin-Borkowski, R. E., Kasama, T., Simpson, E. T., & Feinberg, J. M. (2015). Magnetic Properties of Rocks and Minerals. Treatise on Geophysics, 2nd ed., §2.24. doi:10.1016/B978-0-444-53802-4.00048-8.

Dunlop, D. J., & Özdemir, Ö. (2015). Magnetizations in Rocks and Minerals. Treatise on Geophysics, 2nd ed., §5.08. doi:10.1016/B978-0-444-53802-4.00102-0.

Selected original figure sources

Muxworthy (2013); Muxworthy & Williams (2006, 2009); Muxworthy et al. (2013); Halgedahl & Fuller (1983); Harrison et al. (2002); original Bitter, Kerr, and MFM studies compiled in Roberts Fig. 3.34.

Titanomagnetite composition changes terms in the energy balance



Lower Ms and Tc with Ti

Reduced spontaneous magnetization weakens demagnetizing energy and moves the SD/vortex/MD boundaries to larger sizes. High-Ti phases also lose ordering at much lower temperature.

Easy axes can change

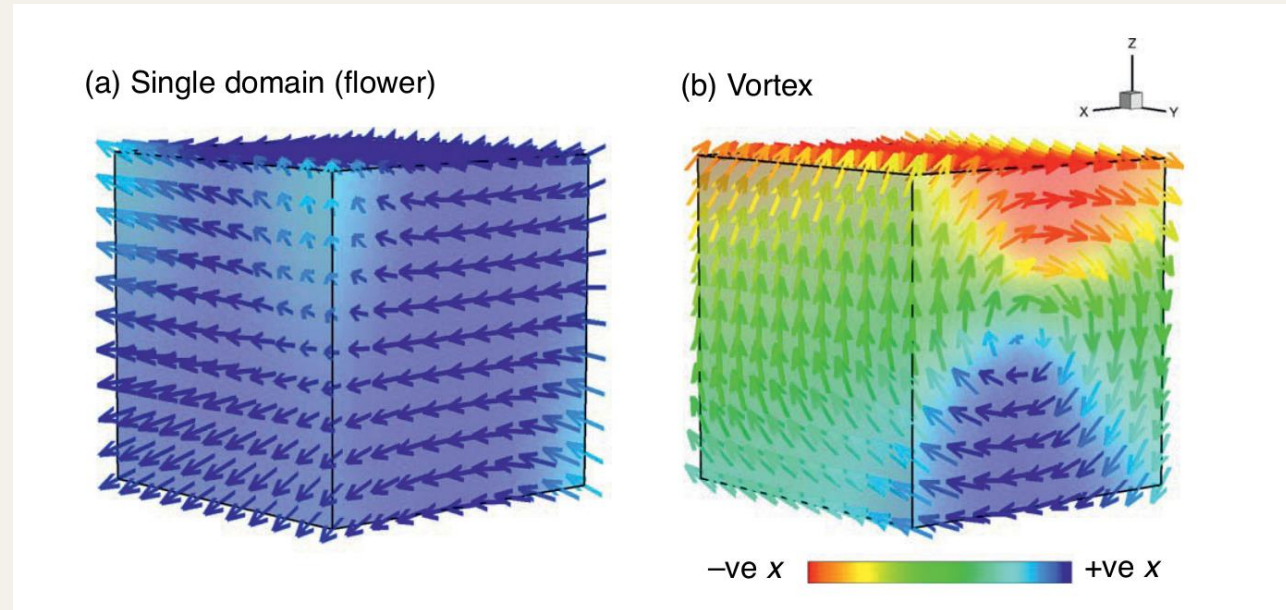
Low-Ti titanomagnetite resembles magnetite with $\langle 111 \rangle$ easy directions; high-Ti compositions can favor $\langle 100 \rangle$, changing allowed wall angles and closure geometry.

Stress matters more

TM60 has strong magnetostriction. Polishing strain, exsolution strain, and defects can dominate coercivity and make domain patterns unusually complex.

Titanomagnetite is just not magnetite with a rescaled grain-size axis.

Greigite supports the same flower-to-vortex topology as magnetite



83 nm cube

Flower state: mostly aligned moments with edge deflection.

107 nm cube

Vortex state: curling closes flux around a core.

Topology transfers across soft ferrimagnets, but critical sizes do not: M_s , anisotropy, shape, and chemistry must be recalculated.

Pyrrhotite, goethite, and hemoilmenite emphasize different controls

Monoclinic pyrrhotite

Lower M_s than magnetite but strong in-plane crystallographic anisotropy. Domain patterns are strongly orientation- and composition-dependent. Room-temperature remanence can be stable; heating risks oxidation or decomposition to magnetite.

Goethite

Weak ferromagnetism plus very high anisotropy and coercivity. Needle shape often matters. Its low ordering temperature and dehydration to hematite complicate thermal work; a magnetite-style vortex interval is not the default model.

Hemoilmenite

Exchange-coupled exsolution lamellae and interface imbalance can dominate the moment. The magnetic recording unit may be a nanoscale lamellar composite rather than a crystallographic grain or classical domain.

Ask first which energy and length scale dominate in that mineral system.

The same grain size can imply a different domain state in each mineral

Mineral	Dominant contrast	Approximate domain-state guide	Interpretive caution
Magnetite	High Ms; modest K	SD ~50-100 nm; broad vortex interval	Shape + interactions shift boundaries
TM60	Lower Ms; high magnetostriction	SD upper guide ~0.2-0.6 μm	Stress, oxidation, exsolution
Maghemite / Ti-maghemite	Vacancies + strain	Nonuniform states likely; d0 varies widely	Thermal inversion and pinning
Greigite	Soft ferrimagnet	Flower and vortex modeled near 0.1 μm	Chemistry and morphology poorly constrained
Hematite	Very low Ms; strong anisotropy	SD upper guide ~15 μm	MD hematite can remain stable
Pyrrhotite	Strong planar anisotropy	SD upper guide ~1.6 μm	Composition and heating reactions
Goethite / hemoilmenite	Anisotropy or interfaces dominate	No simple magnetite-like cutoff	Relevant unit may be needle or lamella

Thresholds are idealized, equidimensional, isolated, room-temperature estimates — not mineral constants.

Sources: Roberts (2025), Chapter 3; Dunlop & Özdemir (2015), Table 3 and §5.08.3. Values are approximate guides.